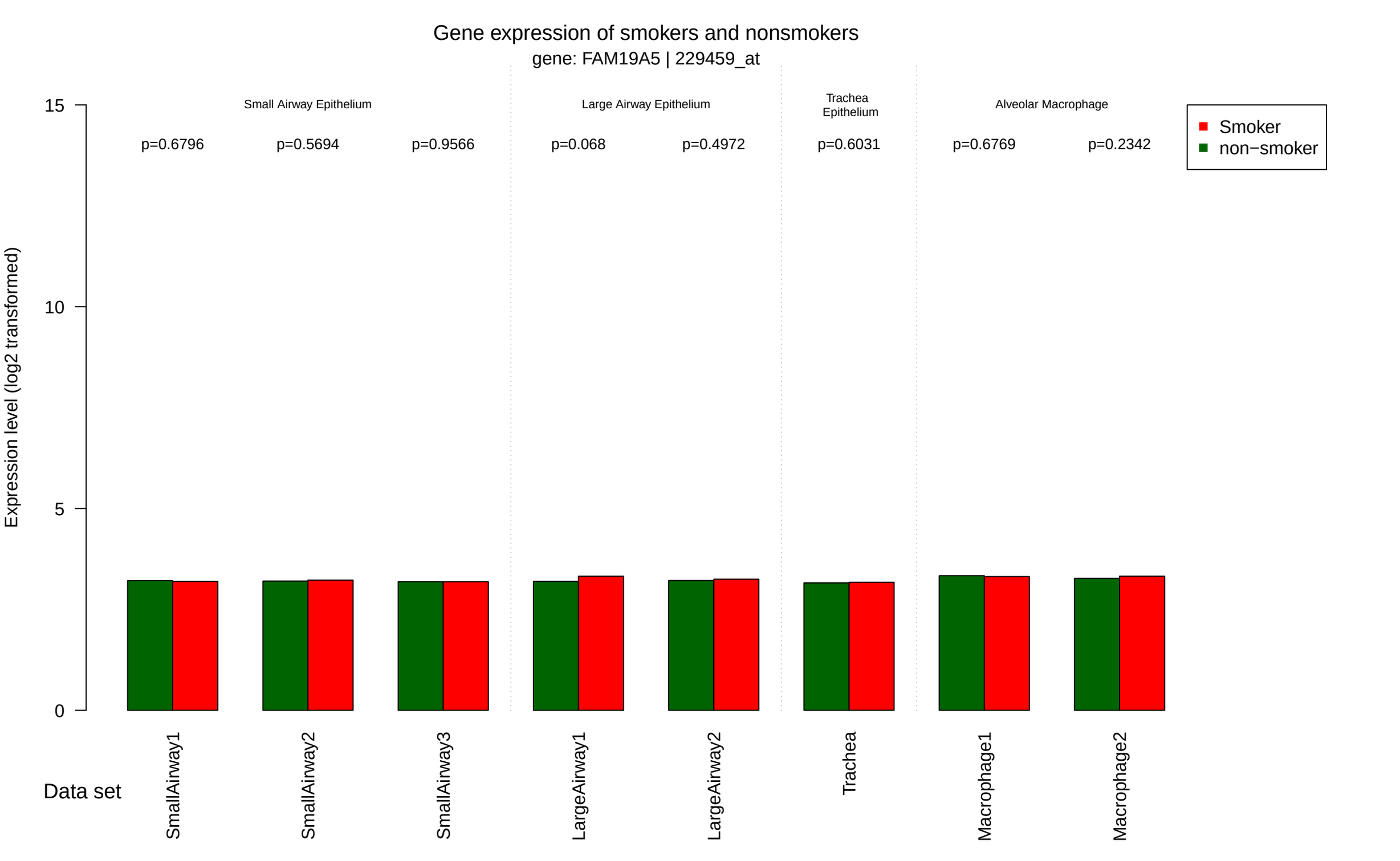
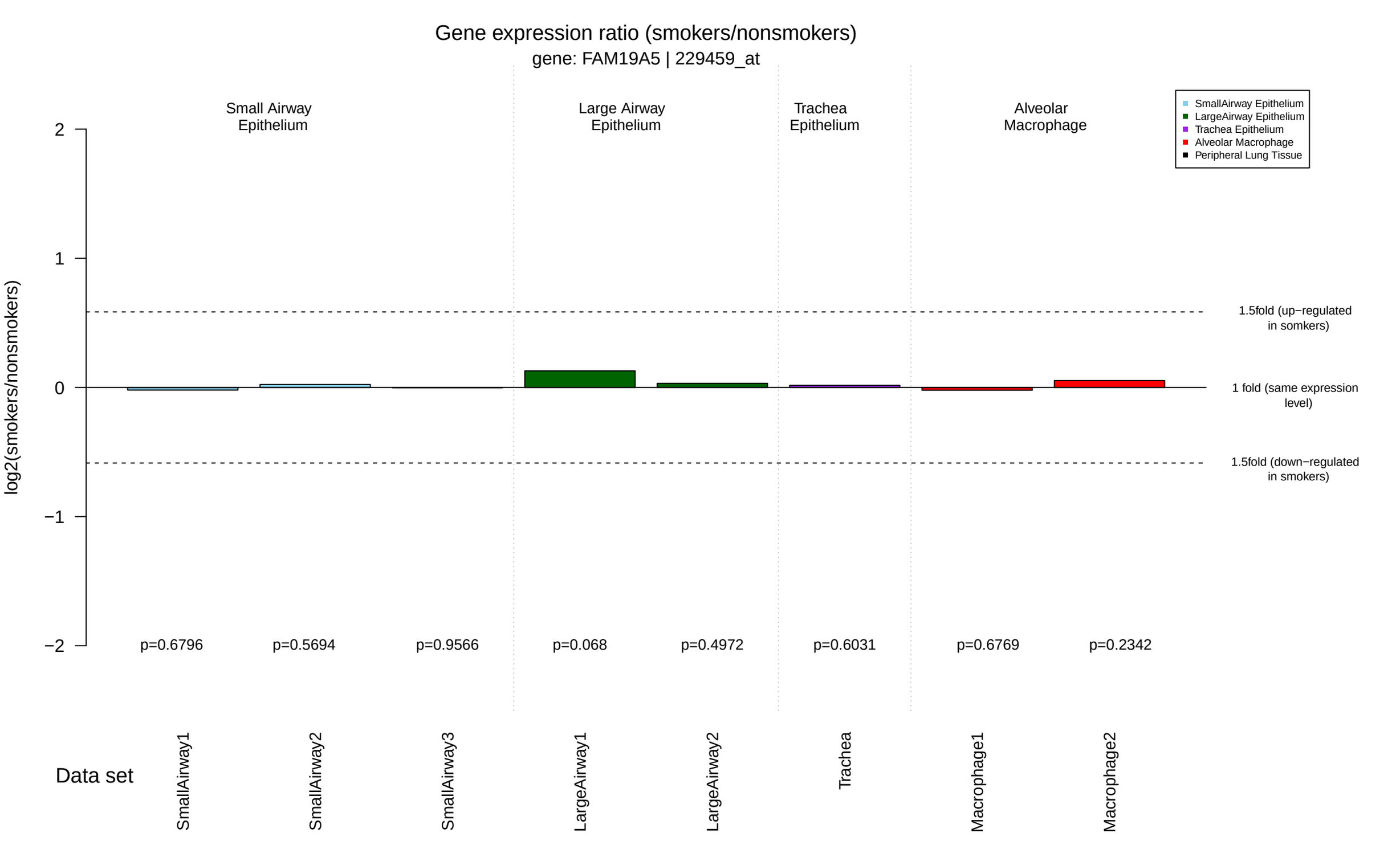
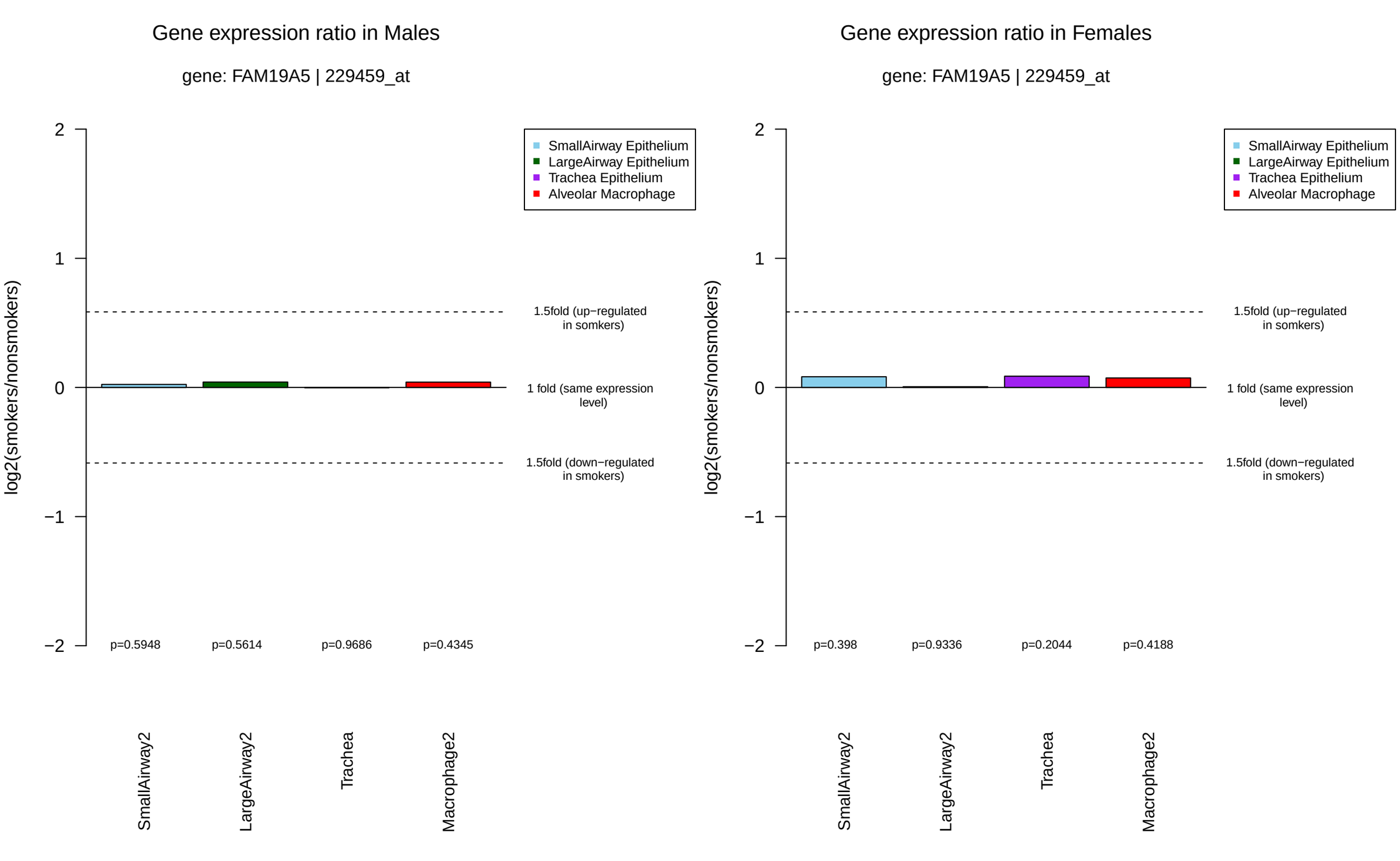
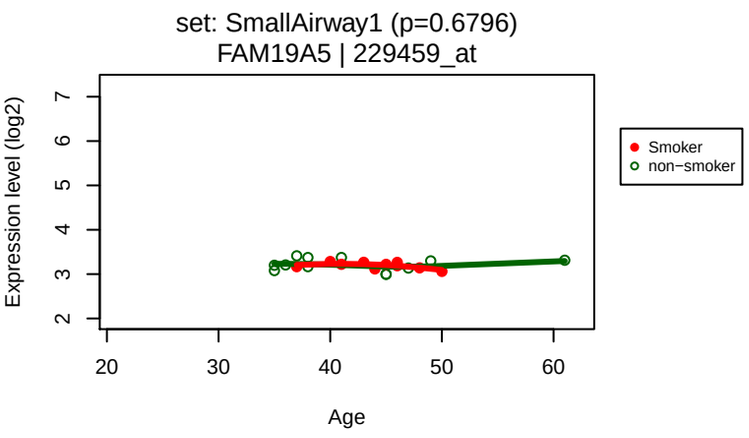
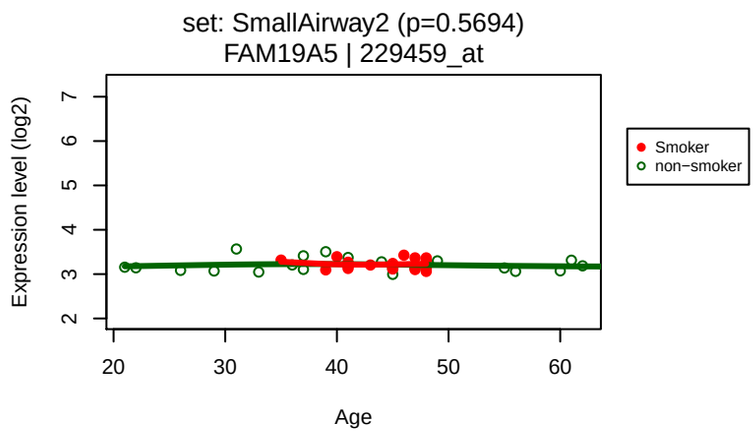
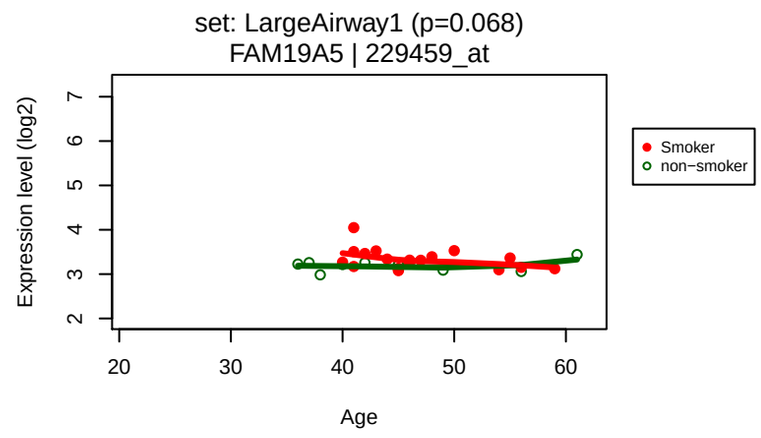

Gene expression of smokers and nonsmokers
gene: FAM19A5 | 229459_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6796
p=0.5694
p=0.9566
p=0.068
p=0.4972
p=0.6031
p=0.6769
p=0.2342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

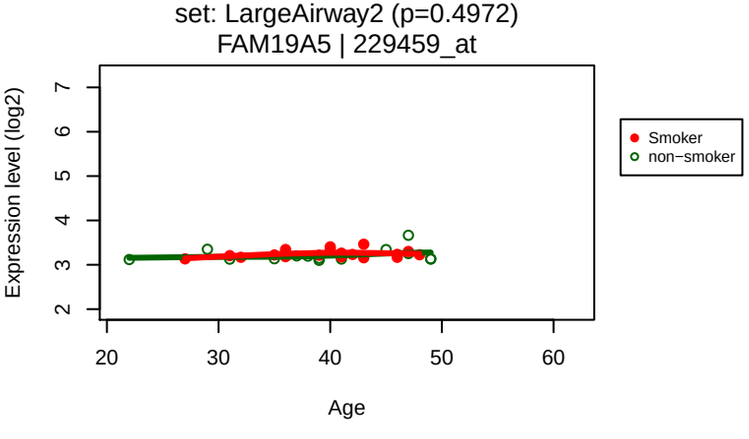

Gene expression ratio (smokers/nonsmokers)
gene: FAM19A5 | 229459_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6796
p=0.5694
p=0.9566
p=0.068
p=0.4972
p=0.6031
p=0.6769
p=0.2342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

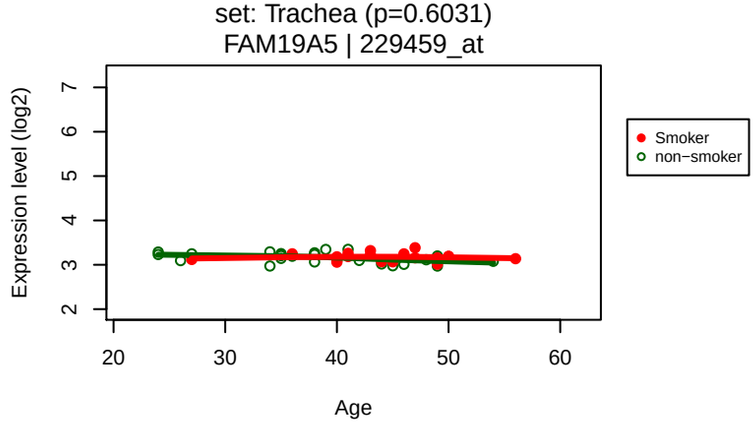

Gene expression ratio in Males
gene: FAM19A5 | 229459_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5948
p=0.5614
p=0.9686
p=0.4345
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FAM19A5 | 229459_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.398
p=0.9336
p=0.2044
p=0.4188
SmallAirway2
LargeAirway2
Trachea
Macrophage2

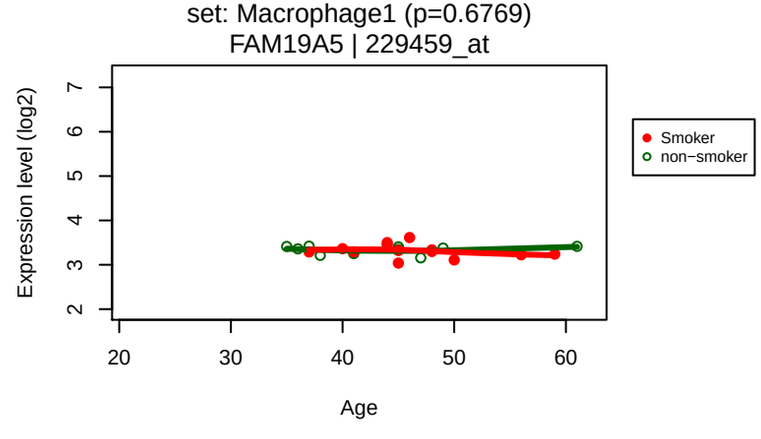

set: SmallAirway1 (p=0.6796)
FAM19A5 | 229459_at
Expression level (log2)
Age
Smoker
non−smoker

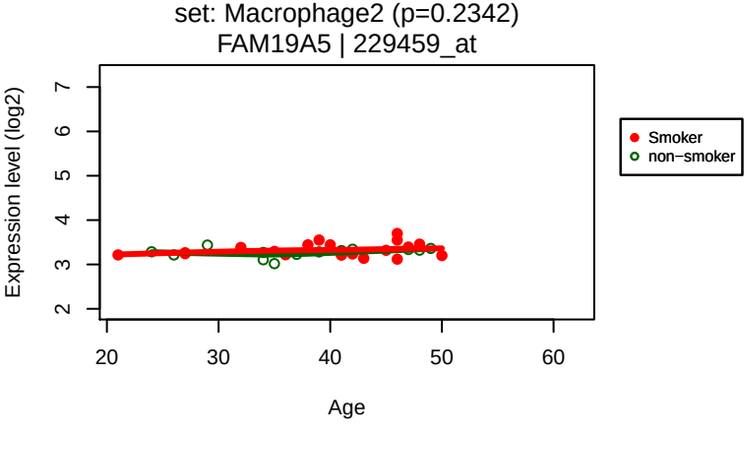

set: SmallAirway2 (p=0.5694)
FAM19A5 | 229459_at
Expression level (log2)
Age
Smoker
non−smoker

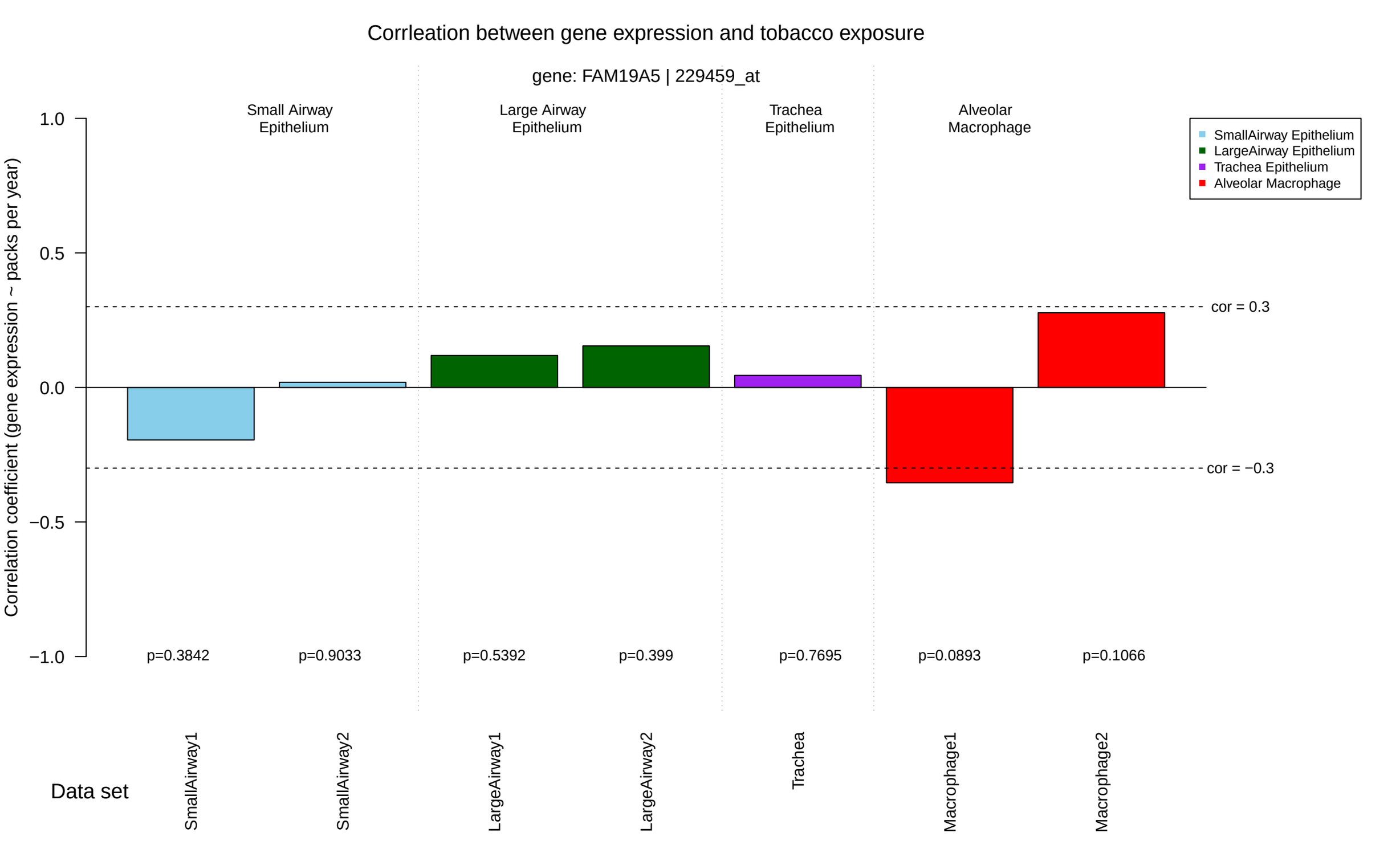

set: LargeAirway1 (p=0.068)
FAM19A5 | 229459_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4972)
FAM19A5 | 229459_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6031)
FAM19A5 | 229459_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6769)
FAM19A5 | 229459_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2342)
FAM19A5 | 229459_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FAM19A5 | 229459_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3842
p=0.9033
p=0.5392
p=0.399
p=0.7695
p=0.0893
p=0.1066
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2